

$2^{(6-2)}$ Fractional Factorial Design

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Introduction

On this report we are going to create a custom set of data(simulation)

Creating the data

We are going to create a custom made linear model with 6 parameters (A,B,C,D,E,F) , just because, I will setup this model to have important A, E, F main effects and EF 2-factor interaction.

```
#Y =      + a*A + e*E + f*F + ef*(E*F) +

set.seed(123)          #

# Σ
mu      <- 30      # I
a_coef  <- 5        # A
e_coef  <- 8        # E
f_coef  <- 6        # F
ef_coef <- -3       # E*F
sigma   <- 2        #      SD

# 0                2^6
A <- rep(c(-1,1), each = 32)
B <- rep(rep(c(-1,1), each = 16), 2)
C <- rep(rep(c(-1,1), each = 8), 4)
D <- rep(rep(c(-1,1), each = 4), 8)
E <- rep(rep(c(-1,1), each = 2), 16)
F <- rep(c(-1,1), times = 32)

# T
```

```
Y <- mu + a_coef*A + e_coef*E + f_coef*F + ef_coef*(E*F) +
  round(rnorm(64, mean = 0, sd = sigma))

Y
```

```
[1] 7 26 33 36 8 29 31 33 7 25 32 37 9 26 29 40 9 22 31 35 6 26 28 35 7
[26] 23 32 36 6 29 31 35 20 38 42 47 19 36 39 45 17 36 37 50 20 34 39 45 20 36
[51] 41 46 18 39 40 49 15 37 40 46 19 35 39 44
```

Running the ANOVA

We will now select only 16 out of these data as if they were in a 2^4 FD, using the generators $I=ABCE=ACDF=BDEF$, and run the ANOVA as usual:

```
Design6 <- data.frame(A=A, B=B, C=C, D=D, E=E, F=F, Y=Y)

# A
# E == A*B*C
# F == A*C*D
Design_FFD <- subset(Design6, E == A*B*C & F == A*C*D)

Y_FFD <- Design_FFD$Y

# T
print(Design_FFD)
```

	A	B	C	D	E	F	Y
1	-1	-1	-1	-1	-1	-1	7
6	-1	-1	-1	1	-1	1	29
12	-1	-1	1	-1	1	1	37
15	-1	-1	1	1	1	-1	29
19	-1	1	-1	-1	1	-1	31
24	-1	1	-1	1	1	1	35
26	-1	1	1	-1	-1	1	23
29	-1	1	1	1	-1	-1	6
36	1	-1	-1	-1	1	1	47
39	1	-1	-1	1	1	-1	39
41	1	-1	1	-1	-1	-1	17
46	1	-1	1	1	-1	1	34
50	1	1	-1	-1	-1	1	36

```
53  1  1 -1  1 -1 -1 18
59  1  1  1 -1  1 -1 40
64  1  1  1  1  1  1 44
```

```
#  $\Phi$       data frame
df_ffd <- data.frame(Y = Y_FFD,
                     A = Design_FFD$A,
                     B = Design_FFD$B,
                     C = Design_FFD$C,
                     D = Design_FFD$D)

#T
model_ffd <- lm(Y ~ A * B * C * D, data = df_ffd)
summary(model_ffd)
```

Call:

```
lm(formula = Y ~ A * B * C * D, data = df_ffd)
```

Residuals:

ALL 16 residuals are 0: no residual degrees of freedom!

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.500	NaN	NaN	NaN
A	4.875	NaN	NaN	NaN
B	-0.375	NaN	NaN	NaN
C	-0.750	NaN	NaN	NaN
D	-0.250	NaN	NaN	NaN
A:B	0.500	NaN	NaN	NaN
A:C	0.125	NaN	NaN	NaN
B:C	-0.125	NaN	NaN	NaN
A:D	-0.375	NaN	NaN	NaN
B:D	-3.125	NaN	NaN	NaN
C:D	-0.250	NaN	NaN	NaN
A:B:C	8.250	NaN	NaN	NaN
A:B:D	0.250	NaN	NaN	NaN
A:C:D	6.125	NaN	NaN	NaN
B:C:D	0.375	NaN	NaN	NaN
A:B:C:D	-0.750	NaN	NaN	NaN

Residual standard error: NaN on 0 degrees of freedom

Multiple R-squared: 1, Adjusted R-squared: NaN

F-statistic: NaN on 15 and 0 DF, p-value: NA

Lenth Test

```
# 1. Get the estimated effects (coefficients) excluding the intercept
effects <- coef(model_ffd)[-1]
factor_names <- names(effects)

# 2. Calculate the Median Effect
ME <- median(abs(effects))

# 3. Calculate the Pseudo Standard Error (PSE)
PSE <- 1.5 * ME

# 4. Compute estimated t-values
tau <- effects / PSE

# 5. Combine into a table for clarity
effects_table <- data.frame(
  Factor = factor_names,
  Estimate = effects,
  PSE = rep(PSE, length(effects)),
  t_value = tau
)

# 6. Sort by absolute effect to easily see the largest
effects_table <- effects_table[order(-abs(effects_table$Estimate)), ]
effects_table
```

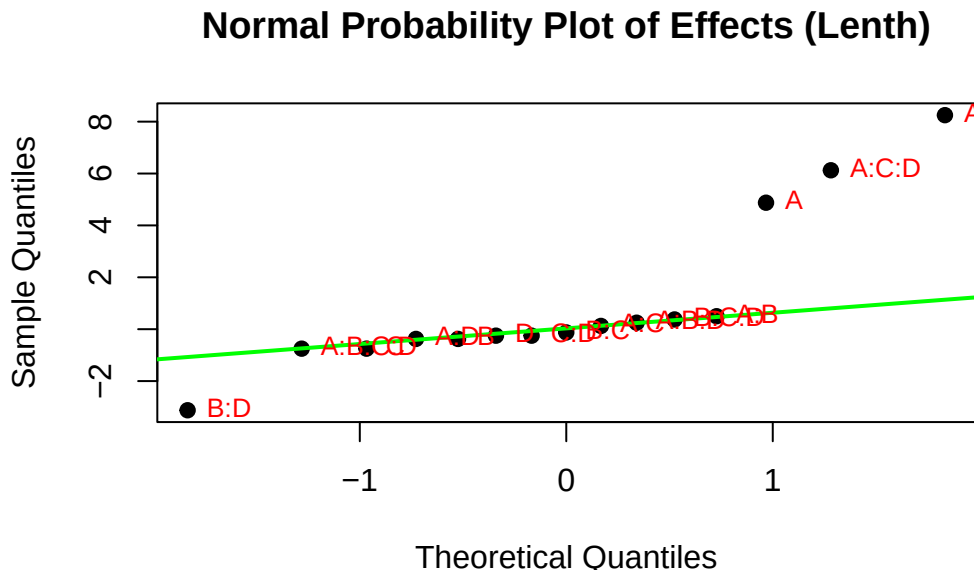
	Factor	Estimate	PSE	t_value
A:B:C	A:B:C	8.250	0.5625	14.6666667
A:C:D	A:C:D	6.125	0.5625	10.8888889
A	A	4.875	0.5625	8.6666667
B:D	B:D	-3.125	0.5625	-5.5555556
A:B:C:D	A:B:C:D	-0.750	0.5625	-1.3333333
C	C	-0.750	0.5625	-1.3333333
A:B	A:B	0.500	0.5625	0.8888889
A:D	A:D	-0.375	0.5625	-0.6666667
B:C:D	B:C:D	0.375	0.5625	0.6666667
B	B	-0.375	0.5625	-0.6666667

A:B:D	A:B:D	0.250	0.5625	0.44444444
D	D	-0.250	0.5625	-0.44444444
C:D	C:D	-0.250	0.5625	-0.44444444
A:C	A:C	0.125	0.5625	0.22222222
B:C	B:C	-0.125	0.5625	-0.22222222

It is clear that $ABC=E$ is important, so is $ACD=F$ and $A \cdot BD$ is shown to be important but we know $BD=EF$ since our generators are $I=ABCE=ACDF=BDEF$.

We finally do the normal probability plot to check the validity of our findings.

```
qq <- qqnorm(effects, main = "Normal Probability Plot of Effects (Lenth)", pch = 19)
qqline(effects, col = "green", lwd = 2)
text(qq$x, qq$y, labels = factor_names, pos = 4, cex = 0.8, col = "red")
```



It is clear that these terms are the ones deviating most from the line. The important terms we found.